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Smart drip irrigation systems using IoT: a review of architectures, machine learning models, and emerging trends

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Abstract

The Internet of Things (IoT) integration in drip irrigation systems is examined in this review paper, with an emphasis on automation, data-driven optimization, and real-time monitoring. The integration of IoT-driven automated drip irrigation represents a transformative leap toward sustainable agriculture, optimizing water conservation, crop productivity, and environmental resilience. Precision farming technologies like artificial intelligence (AI), the Internet of Things (IoT), and 5G-enabled automation are essential for improving irrigation efficiency through real- time monitoring, predictive analytics, and resource optimization as the challenges of climate change and food security worsen. Case studies, including Microsoft's AI-driven initiative in Andhra Pradesh, have demonstrated significant improvements, such as a 30% increase in yield and up to 70% water savings through IoT-based systems. Machine learning techniques like KNN, SVM, ANN, and RF have further refined irrigation scheduling, achieving over 98% accuracy. This review analyzed 56 peer-reviewed studies published between 2015 and 2024, selected based on relevance to IoT-enabled drip irrigation with particular emphasis on literature from the last three years (2022–2024) with AI/ML integration, and deployment aspects. The key research gap identified is the absence of standardized benchmarks, open datasets, and scalable frameworks, which restrict the transition from pilot-scale studies to large-scale real-world adoption. Despite the advantages, barriers such as high initial investment, connectivity limitations, cybersecure- risks, and technical complexity hinder widespread adoption. Addressing these challenges through cost-effective solutions, policy interventions, subsidies, and farmer training programs will be crucial. The evolution toward Industry 5.0, emphasizing human–machine collaboration, AI-powered AgroBots, and energy-efficient smart irrigation solutions, underscores the future of precision agriculture.

Highlights.High-Accuracy Machine Learning Models for Irrigation Optimization Recent studies demonstrate that AI and ML models, particularly Random Forest and Artificial Neural Networks, can optimize irrigation with up to 99.8% accuracy, resulting in water savings of 37% to 70%. However, real-world scalability and adaptability remain ongoing challenges.

AIoT Integration for Real-Time, Precision Agriculture The integration of AI, IoT, and 5G technologies—such as AgroBots, ROS-based irrigation robots, and ZigBee

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mobile irrigation systems—enables real-time crop monitoring, soil assessment, and automated irrigation, showcasing the potential of AIoT-driven smart farming to enhance productivity and sustainability.

Key Research Gaps: Standardization, Adaptability, and Field Validation Despite technological advancements, gaps persist in platform standardization, real-time adaptive irrigation models, and long-term cost–benefit analysis. These limitations hinder large-scale deployment, especially in rural and resource-constrained regions.

Keywords Internet of things (IoT) artificial intelligence (AI) IoT-based smart irrigation, Machine learning in agriculture precision agriculture, Sensor-based irrigation systems AI for water management, Cloud computing in smart farming

1 Introduction

1.1 Drip irrigation in India: enhancing water efficiency and crop productivity amidst challenges

Drip irrigation has emerged as a revolutionary agricultural practice in India, particularly in regions characterized by water scarcity. This technique augments water utilization efficiency, diminishes labor expenditures, and enhances crop productivity, rendering it a viable sustainable solution for a diverse array of crops. The subsequent sections elucidate its ramifications, adoption rates, and effectiveness across various agricultural settings in India. However, despite its advantages, traditional drip irrigation systems are limited by their dependence on manual scheduling, lack of adaptability to fluctuating weather conditions, and inefficient water distribution under heterogeneous soil conditions. These limitations often lead to sub-optimal water use and reduced crop performance, particularly in smallholder farms where resources are constrained. Effects on Water Efficiency and Crop Yield Water consumption is significantly reduced by drip irrigation; empirical studies show that onion growers use 96.6% less water. It enhances crop development and quality, as evidenced by 93.3% of onion growers reporting superior-grade produce [1]. Recent innovations such as IoT-enabled drip irrigation systems integrated with sensors, Arduino microcontrollers, and solar- powered units have significantly enhanced irrigation precision and sustainability, providing real-time monitoring and control of soil moisture and water application [2]. While drip irrigation offers numerous advantages, obstacles such as the need for technical training and financial assistance persist. Addressing these challenges is imperative for realizing its full potential within Indian agriculture.

1.2 Background on smart irrigation and the role of IoT in drip irrigation

In today's world, the Climate crisis- marked by temperature, reduced precipitation, turbulence in rainfall patterns- has interfered with most agriculture activities and activated survival challenges for Indian farmers. Farmers are often not much fully aware about climate change, which ultimately reduces the agricultural yield and leads them to stop opting agriculture as their first income source [3]. IoT-enabled drip irrigation has the potential to overcome many of the shortcomings of conventional drip irrigation. By integrating soil-moisture sensors, climate data, and automated decision-support systems, IoT bridges the gap between static irrigation schedules and dynamic, data-driven water delivery. This makes irrigation more precise, responsive, and sustainable, while also reducing labor costs and the risk of over-irrigation. Global mean temperature is

projected to rise by approximately 2 °C by 2100, primarily due to increased greenhouse gas emissions such as Carbon Dioxide. This warming is expected to alter crop physiology by affecting respiration and increasing the rate of evapotranspiration, thereby intensifying water stress in agricultural systems [4]. Extreme floods and increased frequency of hydrological events are becoming common in India. About 3–6 increase in average temperature increases mean sea level with 15–40% increase in Rainfall in semi-isolated landforms and seaside zones [5]. With careful consideration towards the boundaries of IoT-based Agriculture or Smart Agriculture, it still has great importance to reduce the above-mentioned Climate impacts and to tackle the food security problem [6].

According to market analysts, the global smart irrigation market is valued at approximately USD 2.5 billion (2023) and is expected to grow at a compound annual growth rate of around 15%, potentially reaching USD 5.8 billion by 2030. Adoption of IoT-enabled irrigation remains significantly higher among large-scale, well-resourced farms in developed countries, driven by high capital costs, whereas uptake in developing regions like India remains modest, reflecting infrastructure and financial barriers.

Beyond India, several international deployments highlight the adaptability of IoT-based irrigation. For instance, Spain's vineyards have successfully integrated IoT soil sensors to reduce water use by 25%, while in Israel, large-scale drip irrigation combined with cloud analytics has improved water productivity in arid regions. Similarly, U.S. farms adopting IoT-driven precision irrigation reported yield increases of up to 20%. These examples demonstrate that IoT-based drip irrigation has global relevance across diverse agro-climatic zones.

There are many identified digital technologies which can be equipped with Smart or Digital Agriculture and will be helpful for managing resources and improving decision making processes for efficient agricultural practices. These include Remote sensing, Big Data, AI, IoT-technology and Robots [7]. Agriculture is a data-driven domain that leverages information from various sensors, automated equipment, and weather stations [8] Internet of Things (IoT)- enabled drip irrigation systems are crucial for increasing the efficiency and sustainability of agriculture. IoT enhances crop yields, reduces environmental impacts, and enables accurate water management by leveraging real-time data from several sensors. The main contributions of IoT to drip irrigation are described in the following sections. IoT-enabled sensors track temperature, humidity, and soil moisture, providing farmers vital information for well-informed decision- making [9]. Automated systems can adjust irrigation schedules based on real-time conditions, reducing water wastage and preventing over-irrigation[10]. Despite the significant advantages of IoT in drip irrigation, obstacles like setup costs and the requirement for technical know-how could prevent widespread implementation. However, the promise for improved sustainability and efficiency in agriculture continues to be a strong justification for additional IoT technology investment.

1.3 Importance of water conservation and precision farming

With approximately 72% of freshwater allocated to agricultural use, the withdrawal reached around 4000 billion m3 by 2020 [11]. Future water scarcity due to climate change threatens the sustainability of water practices in agriculture [12]. Socio-economic pressure and climate change imposes threat to water sustainability practices in the agriculture field [13]. The integration of drip irrigation with advanced smart

irrigation technologies markedly improves both water efficiency and agricultural yield. This synergistic approach employs Internet of Things (IoT) frameworks and sophisticated sensors to continuously monitor and manage irrigation processes, thus facilitating optimized water utilization and enhanced agricultural productivity. IoT-based drip Irrigation models enable remote surveillance and management of irrigation systems, empowering farmers to modify water allocation according to real-time data obtained from soil moisture sensors. Empirical research demonstrates that IoT-enhanced drip irrigation systems can achieve remarkable distribution uniformity, with coefficients of uniformity exceeding 90% [14].

1.3.1 Environmental and economic benefits

Intelligent drip irrigation systems contribute to reduced water consumption and mitigate environmental reper- cussions, including greenhouse gas emissions, through the optimization of water and fertilizer utilization [15]. The adoption of solar systems further improves sustainability by harnessing renewable energy sources [10]. Enhanced Crop Performance Investigations reveal that smart irrigation can yield substantial enhancements in both crop productivity and quality, as illustrated in tomato cultivation where the optimization of water and fertilizer management resulted in elevated nutrient content and diminished greenhouse gas emissions intensity [15]. Nonetheless, it is important to note that despite the myriad advantages presented by smart drip irrigation systems, challenges such as the initial capital investment and the requisite technical expertise may impede widespread adoption, particularly among smallholder agricultural producers. Using effective and sustainable farming methods, precision farming technologies can assist farmers produce more crops. These technologies may include remote sensing and drone technology, GPS-guided equipment, and IoT devices [16]. Allied technologies and AI can be very helpful for farmers in taking advanced decisions [17].

Despite the technological promise, socio-economic and ethical concerns remain critical. Smallholder farmers often face high upfront costs, limited digital literacy, and lack of institutional support. Additionally, data privacy issues arise as IoT systems transfer sensitive farm-level data to cloud platforms. Addressing these challenges requires supportive government policies, farmer training programs, and scalable low-cost IoT models to ensure equitable adoption.

The present review was conducted by systematically searching Scopus, IEEE Xplore, ScienceDirect, and Web of Science using keywords such as "IoT irrigation," "drip irrigation automation," and "smart farming." Studies published between 2015–2024 were considered. Inclusion criteria required that papers report experimental validation, simulation, or field deployment. Exclusion criteria included purely conceptual frameworks without implementation results. A total of 56 references were shortlisted and critically analyzed for this review.

Looking ahead, the scalability of IoT-based drip irrigation depends on modular architectures, low-power edge devices, and integration with renewable energy sources. Policy-driven subsidies and collaborative platforms will further enable widespread adoption at the farm level, ensuring that smart irrigation contributes significantly to food security and climate resilience.

The comparative assessment in Table 1 indicates that while IoT-based drip irrigation systems offer significant water efficiency and productivity gains, their adoption is

Table 1 Benefits and Challenges of IoT-Integrated Drip Irrigation

Category	Benefits	Challenges	Citations
Water eff	Optimized use, less wastage	High setup cost	[14]
Crop yield	Higher yield, better quality	Needs skills	[15]
Env. impact	Lowers GHG emissions	Low adoption	[10]
Economic	Lower labor, more revenue	IoT infra cost	[16]
Sustain	Solar-powered	Policy support needed	[10]

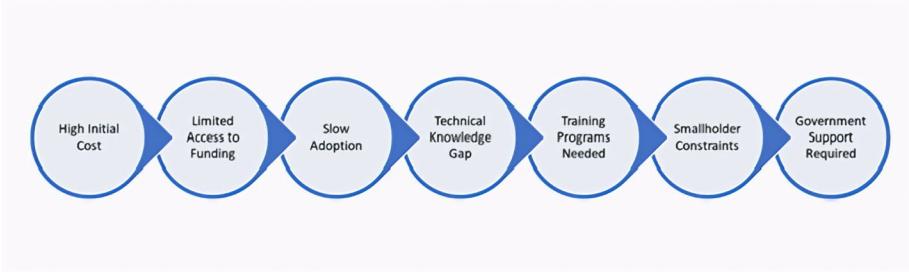


Fig. 1 Challenges in Automated Drip Irrigation Adoption

hindered primarily by economic and technical barriers. For instance, although water wastage can be minimized through optimized scheduling (Suhardi et al., [14]), the high initial setup cost remains a persistent bottleneck for small and marginal farmers.

Similarly, the potential to enhance crop yield and improve quality (Ma et al., [15]) often requires farmers to acquire new technical skills for system calibration and maintenance, which limits widespread diffusion in regions with low digital literacy. The environmental benefits, particularly in terms of reducing greenhouse gas emissions, are well documented (Yadav et al. [10]). However, these advantages are offset by the slow rate of adoption, suggesting the need for awareness programs and incentive schemes.

From an economic perspective, IoT systems can reduce labor dependency and generate higher revenues (Getahun et al., [16]). However, the associated infrastructure and maintenance costs remain challenging in low-resource settings. Lastly, while solar-powered solutions improve sustainability, their scale-up requires supportive policy interventions (Yadav et al., [10]).

IoT-enabled drip irrigation relies on decision-making logic that integrates sensor input (soil moisture, temperature, humidity, and crop type) with predefined thresholds or machine learning models. The decision framework typically involves: (a) continuous monitoring of soil and climate conditions, (b) comparison with optimal thresholds, and (c) automated actuation of valves or pumps. A decision logic diagram 4 illustrates this process, ensuring transparency in how water distribution is optimized. Such frameworks make the system adaptive, reducing human dependency while improving reliability.

In summary, the benefits and challenges are not isolated; they are interconnected trade-offs. High efficiency and yield potential are often constrained by financial and policy gaps, highlighting the need for integrated strategies that combine technological innovation with institutional support Fig. 1.

2 IoT architecture for drip irrigation

2.1 Components of an IoT-based drip irrigation system

Several interconnected parts make up an Internet of Things (IoT)-based drip irrigation system, which allows for automated water use optimization, real-time monitoring, and more.

The key components include:

Fig. 2

2.1.1 Sensors

These gather soil and environmental data in real time. Today, manufacturers and farmers are interested in using inexpensive sensors that can be further connected to nodes to build an affordable solution for IoT-based irrigation systems and smart irrigation. Numerous sensor types, such as pH, moisture, humidity, proximity and water flow sensors, can be incorporated into a basic Internet of Things system architecture [18]. With the right IoT, these inexpensive sensors can help with accurate field moisture content control and water application in plant root zones [19]. According to Mohammed et al., [20], sensor-based irrigation scheduling for automated irrigation is significantly more effective than time-based scheduling. IoT-based irrigation relies heavily on sensors because they transform any physical quantity into an electric signal that the controller can then read [21]. Examples of these sensors include temperature sensors, humidity sensors, soil-moisture sensors, pH sensors, light-dependent resistor (LDR) sensors, rainfall sensors, and NPK sensors Fig. 3.

2.1.2 Microcontroller/microprocessor

This particular unit Controls and processes data from sensors. Sensors send data to Microcontroller. The pillar of a whole IoT irrigation is based on a Smart Microcontroller which further controls the water pump (switching ON/OFF). The selection criteria again depend on the types of IoT irrigation system needed, types of crops, irrigation needs [22]. Common choices are Arduino, ESP8266, ESP32, Raspberry Pi.

2.1.3 Actuators

Solenoid valves motorized pumps used to regulate water flow based on sensor data. Smart irrigation systems have recently evolved, building upon IoT technologies and employing state-of-the-art sensors, actuators, processors, etc. to measure or infer various soil, weather or crop parameters and support decisions regarding plant irrigation [23]. There is alternative to choose IoT based solar pumps which are more economical [24].

2.1.4 Communication module

The use of wires in communication among nodes is problematic and expensive too. So the wireless solutions for communications like Wi-Fi (ESP8266, ESP32), LoRa, Zigbee,

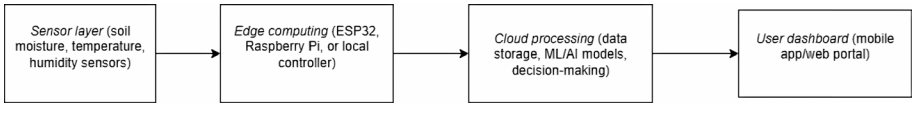


Fig. 2 Proposed IoT-based Smart Irrigation System (high-resolution)

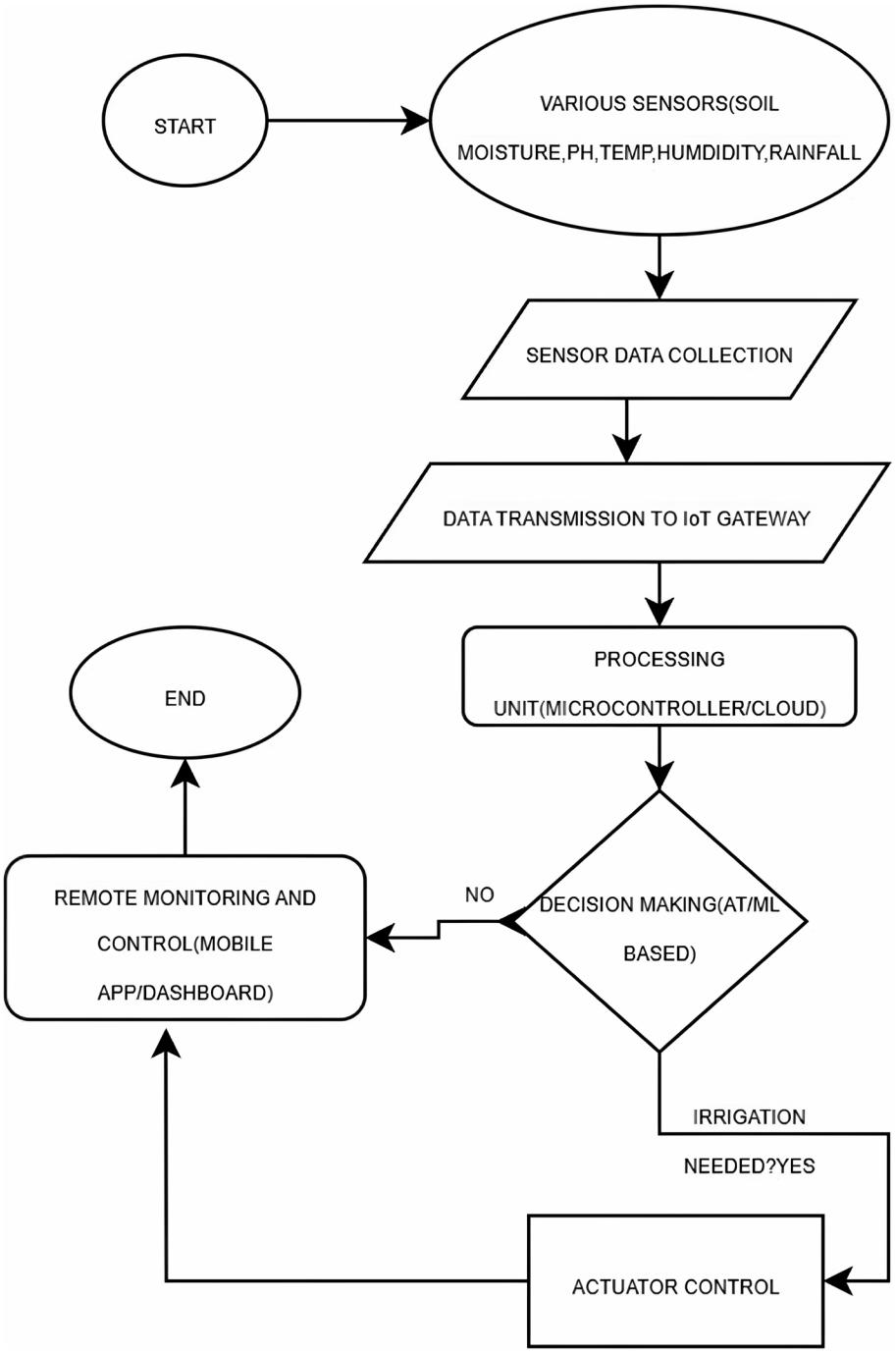


Fig. 3 Schematic Procedural Diagram of an IoT-Based Drip Irrigation System

GSM/GPRS, or Bluetooth provide economical solutions [25]. LoRA is a radio communication protocol (Long-Range). It utilizes spread spectrum modulation technique [26].

2.1.5 Cloud platform/edge computing

Cloud platforms (such as AWS IoT, Google Cloud IoT, Microsoft Azure IoT, ThingsBoard) store, analyze, and process data. In reality, edge computing is the expansion of network computing services from the original centralized cloud-based station. Sensor-generated data can be first gathered and examined before being transmitted to the

central cloud unit for machine learning algorithms to extract additional crucial information [27]. In a number of industries, including healthcare, transportation, and smart irrigation, the convergence of cloud computing, edge computing, and the Internet of Things has greatly decreased latency and bandwidth [28] Fig. 4.

2.1.6 Mobile app/web dashboard

The integration of mobile applications and web dashboards in IoT-based drip irrigation systems significantly enhances agricultural efficiency and resource management. These platforms allow farmers to monitor and control irrigation processes remotely, utilizing real-time data from various sensors. This capability not only optimizes water usage but also supports sustainable farming practices. User interface for farmers to monitor and control irrigation remotely. Provides real-time alerts, analytics, and automation settings. The modern agricultural challenges like productivity, sustainability, resource efficiency needs some deep insights of collected Data. An AI-driven app can help with that [29]. Applications like Blynk provide customizable dashboards to visualize sensor data, enabling users to track soil conditions and water levels easily [30].

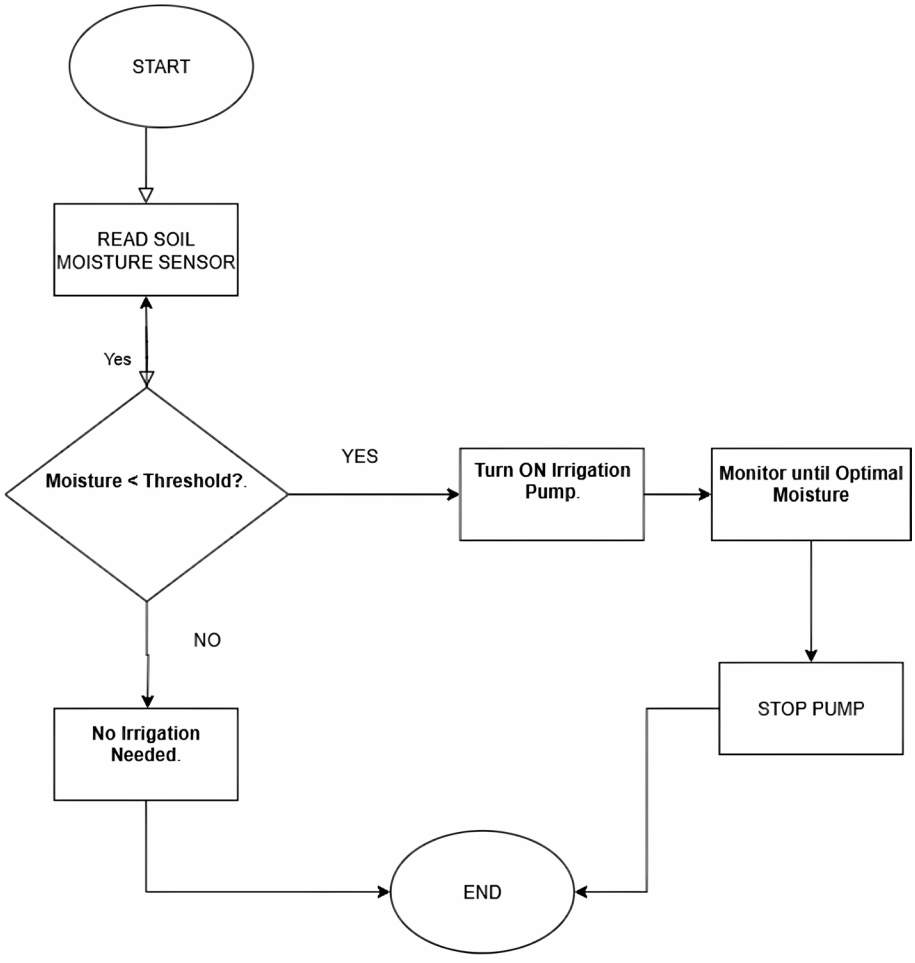


Fig. 4 Control and Decision Logic for the IoT-Based Smart Irrigation System

2.1.7 Power supply

The power supply has a direct impact on the sustainability and effectiveness of IoT-based smart irrigation systems, making it an essential component. Numerous strategies have been investigated to guarantee a consistent supply of power, especially using renewable energy sources like solar power, which improves the system's independence from traditional electricity and lowers carbon emissions. The power supply techniques used in smart irrigation systems are described in depth in the sections that follow. In order to ensure continuous operation without dependency on grid electricity, many smart irrigation systems use solar panels to power sensors and microcontrollers [2]. Solar energy systems show promise as economical and practical ways to reduce fuel consumption in agricultural practices [24].

2.1.8 Use of AI/ML models in smart irrigation systems

AI-based models optimize irrigation schedules using weather forecasts, soil conditions, and crop type data. Advanced machine learning models, including K-Nearest Neighbors and Decision Trees, analyze this data to predict optimal irrigation schedules, ensuring precise water application [31].

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) have significantly shaped the design of IoT-enabled smart irrigation systems. However, deploying ML/DL models in agricultural fields requires careful consideration of computational constraints, energy usage, and real-time decision making Table 2.

Lightweight algorithms such as K-Nearest Neighbors (KNN), Logistic Regression (LR), Naïve Bayes (NB), and Support Vector Machines (SVM) are widely adopted in irrigation systems due to their low computational overhead, faster inference, and minimal energy consumption. For instance, Ndunagu et al. [18] and Jeevanantham et al. [42] used soil moisture, humidity, and temperature sensors connected to NodeMCU/Arduino to run KNN and SVM classifiers locally, achieving greater accuracy 90% with negligible latency. Similarly, Manikandan et al. [43]. reported 42% water savings using a hybrid K-means + SVM model, which is computationally efficient and suitable for real-time control on microcontrollers.

In contrast, deep learning (DL) approaches such as artificial neural networks (ANN), convolutional neural networks (CNN), and long short-term Memory (LSTM) offer superior prediction accuracy but require higher computational power, memory, and energy, often necessitating cloud integration. For example, Amassmir [41] compared KNN, SVM, and ANN for irrigation decision-making, showing ANN achieved the highest precision (96.81%) but required a Raspberry Pi 3 for processing. Similarly, emerging studies explore reinforcement learning for dynamic water scheduling, though its computational demand restricts in-field adoption without edge-cloud partitioning.

Table 2 Comparison of Smart Irrigation Systems

Feature	Traditional Irrigation	IoT-based drip irrigation
Water efficiency	50–60%	90–95%
Energy consumption	High	Lower (solarpowered available)
Automation	Manual	Automated & AI optimized
Cost (initial investment)	Low	Moderate–high
Long-term savings	Limited	Significant (water conservation)
Scalability	Limited	Highly scalable

Latency: Lightweight models can provide near-instant predictions (< 1 s) on microcontrollers, while DL models introduce higher delays if run locally or depend on cloud connectivity.

Energy Usage: Frequent sensor data processing with DL models consumes significantly more energy, which is a critical issue in solar/battery-powered IoT systems.

Real-Time Applicability: Hybrid approaches (ontology + ML or cloud + edge processing) are gaining attention. For instance, a hybrid irrigation system reported by Kaur et al. [46]. used RF and NB on cloud platforms, achieving 99.8% accuracy with low mean squared error (MSE = 0.008), while maintaining efficient in-field execution.

With the advent of AI for agricultural Innovation (AI4AI), cognitive computing and cloud-assisted ML are expected to drive disruptive changes. However, for resource-constrained IoT environments, lightweight ML remains more practical, whereas DL can be selectively integrated at the cloud/edge layer to enhance prediction accuracy without compromising field-level responsiveness. A balanced adoption of lightweight models for field decision-making and complex models for long-term analytics may represent the optimal design paradigm.

3 Advantages and challenges of IoT-based drip irrigation

The integration of IoT with agricultural activities has advanced with recent technological developments that bring convenience to our fingertips. The variety of applications lies in food production, agriculture, self-sustaining operations, mobility, minimal upkeep, efficiency, robust design, and dependability. The agricultural sector is ever evolving. A suitable mobile-friendly software system can help integrate all the agricultural activities (from planting to harvesting) with cloud computations, AI, Big-Data, etc. An improved decision-making approach using an effective machine learning solution can be developed to enhance water use efficiency and ensure sustainability for future applications [32]. There are some challenges that also need to be addressed while using IoT are Enhanced performance, expenditure, uniformity, diversity, availability, flexibility, power efficiency, integration capability, dependability, portability, and climatic factors [33]. The author suggested some other issues such as upfront expenses, information handling, multi-platform compatibility, network accessibility, protection measures, dependability, expansion capability, and compliance issues in his work [21]. Author Idoje et al. [26], concluded in his work that communication protocols that have been used in field are falling due to large distances. Intelligent devices that may be battery-operated can also cause reduced operational efficiency. Data security is another concern with network issues. Incorrect weather detection can also interfere during communication. There is not much research associated with the detection of plant disease. Perception layer devices must withstand harsh conditions and operate long-term on limited power, requiring energy-efficient programming to avoid frequent maintenance. The vast number of interconnected devices generates massive data, soon exceeding small-scale server capacities [34]. Device performance challenges and reliability of operating hours of the entire system are questionable when there is wide heterogeneity created due to the presence of devices, associated operating systems [35] Fig. 5.

Real-world deployment challenges remain a significant barrier to scaling IoT-enabled drip irrigation. Connectivity issues are common in rural areas, where limited network coverage hampers real-time data transfer, especially for cloud- reliant systems. Latency

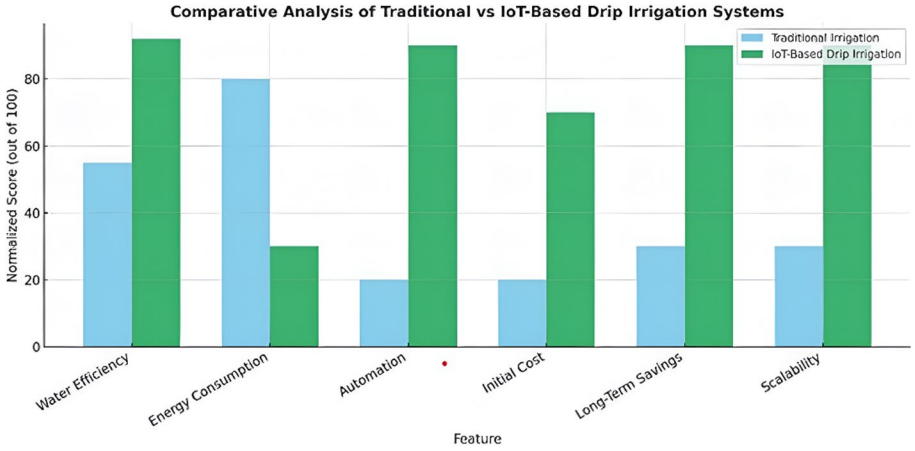


Fig. 5 Comparison of Traditional Irrigation vs IoT-Based Drip Irrigation Systems. The bar chart provides a comparative analysis between traditional irrigation and IoT-based drip irrigation systems across six key parameters: water efficiency, energy consumption, automation, initial cost, long-term savings, and scalability. IoT-based systems outperform traditional methods in efficiency (90–95%), automation, and energy consumption. Despite a higher initial cost, they offer long-term savings and scalability, highlighting a shift towards smart irrigation for sustainable agriculture

in data transmission can delay decision making, reducing the effectiveness of automated irrigation. Moreover, high initial investment costs for hardware (sensors, controllers, communication modules) and maintenance discourage adoption among smallholder farmers. Energy requirements also pose difficulties; although solar-powered systems reduce dependency on grid electricity, their efficiency depends on local climatic conditions. These challenges underscore the need for cost-effective low-power IoT solutions, local edge processing, and robust rural connectivity infrastructure to ensure reliable deployment at scale.

4 Related work

The researcher highlighted the future upcoming explosion in AI for Agricultural Innovation (AI4AI), which includes the Cognitive computing technique growing as the most disruptive technologies. Microsoft is working with 175 farmers in Andhra-Pradesh just to provide advisory services in agriculture practices. Initial level research already showed higher production 30%. Sensors used in his work are soil humidity sensor, temperature sensor, and rainfall sensor. Data collected using an acquisition map. Data storage is done in Node-RED and MongoDB platform. KNN, Logistic Regression, Neural Networks, Support Vector Machine, and Naive Bayes are models used for prediction analysis. With a regression rate of 98.3% and an RMSE of 0.12, the K-NN prediction modeling approach was the most effective [35] Table 3.

Researchers described a smart irrigation architecture where decision making is based on ontology and machine learning. The basic ontology technique is used for determining the water needs based on crop type, climate condition, soil type. Real time sensor data is also collected but the final decision of watering plants is based upon 50–50. This hybrid approach is assuring efficient water usage [36].

The potential of smart irrigation systems using IoT, embedded systems and cloud computing can address food production with views of climate change and traditional irrigation methods. The system adopted by the researcher includes the integration of real time data collection from sensors (Moisture, humidity, temperature and water levels using

Table 3 Components, advantages, and challenges of IoT-based drip irrigation

Component	Function	Citations
Sensors	Measure soil moisture, temp, pH, weather	[18]
Microcontroller	Processes sensor data, controls irrigation	[22]
Actuators	Operate valves, pumps for water flow	[23]
Comm. Module	Enables wireless data via LoRa, Wi-Fi, etc	[25]
Cloud platform	Stores, processes real-time irrigation data	[27]
Mobile app	Allows remote irrigation monitoring/control	[29]
Power Supply	Supports solar/battery for off-grid use	[24]
AI/ML	Optimizes schedules via weather, crop data	[32]
Security	Protects data via TLS, MQTT, Blockchain	[34]

sensors DHT22, Water level sensors and soil moisture sensors) connected to microcontroller ESP32. They opted for systems with the ThingSpeak cloud and ThingView app for wireless data transmission which enables control of pumps and remote monitoring. The results demonstrate a significant 70% reduction in water consumption for soil irrigation [37].

Several IoT-enabled irrigation systems have been developed, integrating sensors, microcontrollers, and cloud platforms to optimize water use. One study employed a solenoid valve, GSM module, PIC 16F877A microcontroller, a temperature sensor (LM35 5 V), a soil moisture sensor, and a 16 × 2 LCD to display environmental conditions [38].

Another developed system talks about usage of sensors and IoT based weather data, the experiment was conducted on Mustard leaf cultivation with data driven modeling. The predictive modeling developed to analyze water flow, water loss, soil moisture. Sensors used are Moisture sensor (VH400), IoT based weather station (calculates ETo (Evapotranspiration)) to estimate crop water used and Flow meter (YF-S201 for checking values for amount of water used). The ESPresso Lite V2.0 and Raspberry Pi 3 process the collected data. All data (moisture, weather, and water usage) is sent to the Raspberry Pi for decision-making based on an irrigation scheduling algorithm. A water pump (actuator) is connected to the Raspberry Pi through a solid-state relay to regulate irrigation. A camera connected to the Raspberry Pi captures plant images and uploads them to an IoT webpage every hour for remote monitoring. The data is stored in a database and displayed on an agro-IoT website for farmers to track irrigation trends. The system decides when to irrigate by checking:ETo value (if it increases, more water is released to compensate for water loss) and Soil moisture levels (ensuring soil moisture does not exceed its field capacity) [39].

A different IoT-enabled drip irrigation system combines temperature, humidity, and soil moisture sensors with an Arduino microprocessor. Based on the moisture levels in real time, the Arduino microcontroller regulates a water pump and solenoid valve. A DC power supply supports data communication, while the centralized microcontroller processes sensor inputs and operates the solenoid valve accordingly. The system uses NodeMCU ESP8266 for Wi-Fi connectivity, enabling remote monitoring and control via an IoT cloud platform (ThinkSpeak) and an Android mobile application. The Arduino YUN microcontroller manages sensor commands and network communication, ensuring the pump operates only when necessary, conserving water. A buzzer alerts users in case of critical failures, while a web-based dashboard allows secure access with authentication. This automated system enhances precision irrigation, optimizing water use,

reducing energy consumption, and enabling remote control for farmers through mobile or web applications [40].

Here Author Amassmir [41] worked in Potato and wheat fields. Although they compared 3 Machine Learning Algorithms K-NN, SVM, ANN to find the best intelligent irrigation systems to work with. Sensors used are again Temperature,Humidity and Soil Moisture which again used to train Machine Algorithms. Data from Sensors directly goes to Arduino. Arduino sends this data to Raspberry Pi 3 which runs ML model analyzing when to turn on the pumps. The accuracy obtained from this ML models are 91% for KNN, 87% for SVM and 96.81% for ANN.

Jeevanantham et al., [42] suggested and worked with an IoT based solution to improve conventional soil irrigation systems with smarter and more efficient ones. The systems include NodeMCU (a Microcontroller) and sensors (Moisture, Humidity, Temperature, Rainfall), Wireless Sensor Network to transfer Real-time data to monitor and control water supply. Microcontroller processes sensor data based on user-defined logic and sends signals to control irrigation. The entire system is coded in C-Language to ensure proper decision-making. The software used is Blynk Software which sends email alerts to check upon the Temperature, Rainfall, Humidity and Water levels. The integrated Iot automatically adjusts water supply reducing water wastages.

Manikandan et al. [43], used a system that integrated IoT Sensors and Cloud Storage. Sensors used are soil moisture sensor and humidity sensor. There is a centralized cloud where data gets stored. CFS algorithm which discards irrelevant data and K-means clustering groups similar data. The machine learning models used to analyze the data are SVM, RF, NB. There were a total of 42,665 Rice crop records where 35000 were used for Training the model and 7665 used for testing. Researchers achieved a Hybrid k-means + SVM model which gave 98.5% accuracy with 42% water savings.

There is research in which the IoT system was actually built on various key components. The used one are ESP 8266 NodeMCU (A Wi-Fi enabled one), DS18B20 Temperature Sensor (measure soil temperature), Soil Moisture Sensor, DS3231 Real Time Clock, SIM800L GSM Module which sends SMS Alerts and make calls and the last Nokia 5110 LCD display. The system boots up and collects soil moisture and temperature data using sensors, then connects to Wi-Fi, creating a hotspot if needed. If the connection fails, it restarts and retries. It downloads weather data, including sunlight, humidity, and rainfall, and if unsuccessful, it restarts again. The collected data is uploaded to ThingSpeak (IoT platform), with the system retrying if the upload fails. The real-time data is displayed on an LCD screen. If the system boots in less than a minute, it sends an SMS and makes a call; if a call is received, it responds with an SMS. It then checks the time and schedules irrigation based on soil moisture and rainfall data. If the soil is dry and no rain is expected, the system activates the water pump. This process repeats every 15 min to ensure optimal irrigation. The collected data is sent to the cloud (ThingSpeak) every 15 min, where a machine learning model analyzes it to predict irrigation needs and automatically control watering. The system is programmed using Arduino IDE for microcontroller control, while Python and Jupyter Notebook handle data analysis. ThingSpeak is used for IoT data storage and visualization, and a web interface is developed using HTML, CSS, and JavaScript. Anaconda facilitates Python-based data processing. The system makes smart decisions by automatically controlling the water pump based on soil moisture, temperature, and rainfall data, with a web application enabling

remote monitoring and control. All components are integrated onto a PCB board, and a machine learning model optimizes water use for crops. Users can receive SMS alerts and remotely monitor the system [18].

A smart irrigation system comprising soil moisture and humidity sensors, a Raspberry Pi, cloud storage, a soil dataset, machine learning technologies, and a mobile application was employed in one study by Phasinam et al., [44]. The Raspberry Pi serves as the core processor unit and connects to cloud storage and sensors. DHT11 and DHT22 Humidity sensors record the humidity and moisture content of the soil and send the information to the cloud. YL-69 The water content of the soil was assessed with a soil moisture sensor. The final one, cloud storage, uses machine learning algorithms (SVM, RF, and NB) to store soil and climatic data and provide users with real-time insights through a mobile application. The accuracy and effectiveness of several machine learning methods were examined using a data set of 330 soil recordings. Best results were obtained using SVM algorithms in terms of accuracy (88%) and water savings (37%).

Ullah et al. [45], opted for an Energy-Efficient Water Management Platform that integrates sensors, computational devices, and IoT services for optimized irrigation. Field sensors (Soil-Moisture, Temperature, Water-level Sensors collect Data). The in-field sink is a key component in the wireless sensor network (WSN) deployed within the agricultural field which collects sensor data and transmits it to reduce energy usage. The outer sink serves as a second- level data collection and processing hub, located outside the immediate field area but within the communication range of multiple in-field sinks. It filters redundant data to optimize communication and energy consumption. IoT service cloud manages data, device registration and user notifications. There are valve controllers and IoT-enabled valves which automate water flow based on real time conditions. Sensors detect Soil Moisture, temperature and water levels. These collected data from sensors processed by the fusion center and transmitted to the IoT cloud. There are the predefined threshold values based on which automated valves operates (SOIL MOISTURE: VALVE OPENS WHEN SOIL MOISTURE IS LESS THAN 10% AND CLOSES IF SOIL MOISTURE IS GREATER THAN 60%). The whole system was tested against the SWAMP Model using MATLAB 2019a. About 100 sensor nodes were deployed when in a 400400 m field. The entire work shows that EEWMP consumes less energy than SWAMP extending network lifespan. After 1100 s, SWAMP is exhausted while EEWMP retains 30% energy.

Research by Kaur et al. [46] in Sri Ganganagar, Rajasthan, suggests a clever hybrid irrigation system that uses climatic variables and soil moisture monitoring to automate watering. Multiple sensors (soil moisture, wind speed, water level, rain detection, temperature, and humidity sensors) are integrated into the system, which is primarily controlled by a NodeMCU (ESP8266).Farmers can remotely monitor and manage irrigation thanks to these sensors' real-time data collection and transmission to the ThingSpeak cloud server and Blynk application.In order to forecast irrigation demand based on soil moisture, temperature, humidity, and motor condition, this study assesses five machine learning (ML) algorithms: KNN, Naïve Bayes (NB), Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM). A 3000-recording dataset was split among 20% testing and 80% training. RF NB had the lowest mean square error (MSE = 0.16) and the highest accuracy (99.8%). LR's accuracy was 99.5% (MSE = 0.5). With corresponding MSE values of 0.66 and 1.66, SVM and KNN demonstrated 99.3%

accuracy. Testing and Training: To find the most accurate method for smart irrigation decision-making, the data was processed using the ThingSpeak cloud and then examined using machine learning models.

To ensure objectivity in the literature review, a systematic approach was adopted. Peer-reviewed journal articles, conference proceedings, and high-impact reports published between 2010 and 2024 were considered. Databases such as Scopus, Web of Science, and Google Scholar were utilized to retrieve studies using keywords including "IoT in irrigation," "smart farming," "precision agriculture," and "data-driven water management." The inclusion criteria were: (i) studies focusing on IoT or data-driven applications in irrigation, (ii) reported outcomes in terms of water savings, efficiency, or yield improvement, and (iii) availability of technical or experimental details. Exclusion criteria included non-peer-reviewed sources, non-English publications, and studies lacking sufficient methodological description. This structured filtering ensured both breadth and rigor in the reviewed literature Table 4.

4.1 Methodological differences in IoT-based drip irrigation

While Table 5 provides a comparative overview of sensors, models, and outcomes, it is equally important to distinguish the methodological approaches:

Rule-Based Approaches: These rely on predefined thresholds (e.g., soil moisture < 30% → irrigation ON). They are simple, cost-effective, and require minimal computation but often fail to adapt under changing environmental conditions.

ML-Based Approaches: Machine learning models (e.g., Random Forest, ANN, CNN) leverage sensor data and historical datasets to predict irrigation needs more dynamically. These approaches improve accuracy and adaptability but require large datasets and computational resources.

Hybrid Approaches: Some studies combine rule-based thresholds with ML predictions, ensuring robustness even when data is sparse. These offer a balance between practical deployment and predictive intelligence, often leading to higher water savings.

Table 4 Potential Aspects and Insights

Aspect	Key insights	References
Role of 5G and AIoT in smart farming	AI and IoT help in soil management, crop health monitoring, disease detection, and weed control. 5G improves data transmission speed, low latency, and connectivity for precision agriculture	Sharma & Shivandu [48], Mohamed et al. [20]
5G benefits	Enables real-time crop monitoring, predictive analytics, and food traceability with high-speed data trans- mission. Supports WSNs, UAVs, and IoT sensors for efficient irrigation	Boursianis et al. [50]
Automation in irrigation	Robots equipped with sensors and actuators improve irrigation efficiency. They can be solar-powered for sustainability	Obaideen et al. [51], García et al. [52]
Smart irrigation robotics	Robots use ZigBee-based wireless systems for irrigation, reducing space consumption and maintenance. Future enhancements include remote sensing and ultrasonic sensors	Bodunde et al. [53]
Industry 5.0 in agriculture	Human–machine collaboration enhances productivity. Robotics automate plowing and sowing while humans oversee operations	Singh & Sobti [54]
Future Role of Robotics	Robotic arms will revolutionize agricultural production, increasing efficiency, innovation, and sustain- ability	Jin & Han [55]

Table 5 Summary of smart irrigation systems: sensors, platforms, ML models, and key outcomes

S. No	Authors & year	Sensors used	Platform	ML models used	Accuracy/outcome
1	Tace et al. [36],	Soil Humidity, Temperature, Rainfall	Node-RED, MongoDB	KNN, LR, NN, SVM, NB	KNN: 98.3%, RMSE = 0.12; 30% higher crop yield
2	Munir et al. [37],	Ontology-based	ML + Ontology	Hybrid Decision Logic	Rule + ML; not quantified
3	Morchid et al. [38],	Moisture, Temp., Water Level (DHT22, YL69)	ESP32, ThingSpeak	Not specified	70% water reduction; wireless pump control
4	Ghodake & Mulani, [39]	LM35, Soil Moisture Sensor	PIC 16F877A	Not specified	Real-time display via fISM
5	Abioye et al. [40],	VH400, Weather Station, Flow Meter	Raspberry Pi, ESPresso Lite	Scheduling Algorithm	IoT webpage + image capture + autopump
6	Jain [41],	Temp., Humidity, Soil Moisture	Arduino, NodeMCU	Not specified	Web + mobile dashboard for control
7	Amassmir et al., [42]	Temp., Humidity, Soil Moisture	Arduino, Raspberry Pi	KNN, SVM, ANN	ANN: 96.81% best; potato/wheat fields
8	Jeevanantham et al., [43]	Temp., Rainfall, Humidity, Moisture	NodeMCU, Blynk	No ML; logic-based	Alerts + wireless sensor network
9	Manikandan et al., [44]	Soil Moisture, Humidity	Cloud Storage	SVM, RF, NB, k-Means + SVM	98.5% accuracy; 42% water saving
10	Ndunagu et al. [18],	Soil Temp., Moisture, fISM, RTC	ESP8266, fISM, Python	ML (unspecified)	IoT UI + SMS alerts + auto control
11	Phasinam et al. [45],	DHT11, DHT22, YL-69	Raspberry Pi, Cloud, App	SVM, RF, NB	SVM: 88%; 37% water saving
12	Ullah et al. [46],	Moisture, Water Level, Temp	WSN, MATLAB	Not specified	EEWMP: 30% more energy saved
13	Kaur et al. [47],	Soil Moisture, Wind, Rain, Temp., Water Level	NodeMCU, Blynk, ThingSpeak	KNN, NB, LR, RF, SVM	RF/NB: 99.8%, SVM: 99.3%; Rajasthan fields

This categorization highlights that while rule-based systems dominate low-cost deployments, ML- and hybrid approaches provide scalability and precision, making them more suitable for large-scale smart irrigation systems.

The statistical findings on water savings and accuracy in the reviewed studies were primarily derived from experimental trials, field validations, and simulation-based analyses. For instance, soil moisture sensors and weather stations were calibrated against ground-truth data to ensure accuracy before large-scale deployment. Many studies employed comparative designs, where IoT-based systems were benchmarked against conventional irrigation practices, with performance indicators such as water use efficiency, yield per unit of water, and irrigation uniformity. Statistical validation typically included error metrics (e.g., RMSE, MAE) and confidence intervals, ensuring the reported outcomes were both reliable and replicable Figs. 6 and 7.

5 Future trends and research directions

5.1 Revolutionizing agriculture with AI, IoT, and 5G: advancing smart farming and precision irrigation

The integration of AI and IoT can benefit the Agricultural field in many different ways. AI tools can be applied to do soil management, crop health monitoring, disease detection and weed control by using remote sensing or automated robots(AgroBots) [47].The development of smart systems has led to a considerable evolution in communication technology. IoT-based smart agriculture was supported by 3G/4G/NB-IoT networks, although 4G efficiency has decreased due to rising data demands. Nearly 100 times faster

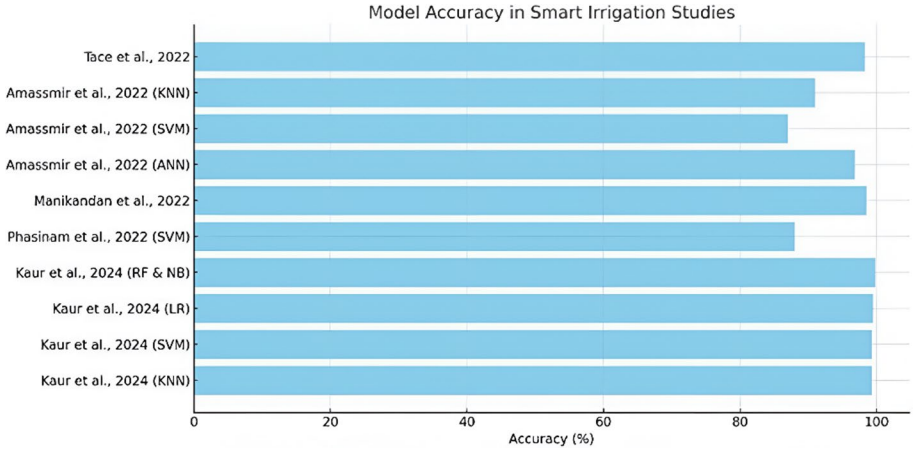


Fig. 6 Comparison of Prediction Accuracy Across Different Machine Learning Models in Smart Irrigation Studies. The bar graph shows that Kaur et al. [47]. achieved the highest prediction accuracy using Random Forest (RF) and Naive Bayes (NB), both at 99.8%, followed by Logistic Regression (99.5%), SVM (99.3%), and KNN (99.3%). Tace et al. [36]. and Manikandan et al. [44]. also reported high performance (> 98%). Phasinam et al. [45]. reported a lower accuracy of 88%, suggesting scope for improvement in model optimization

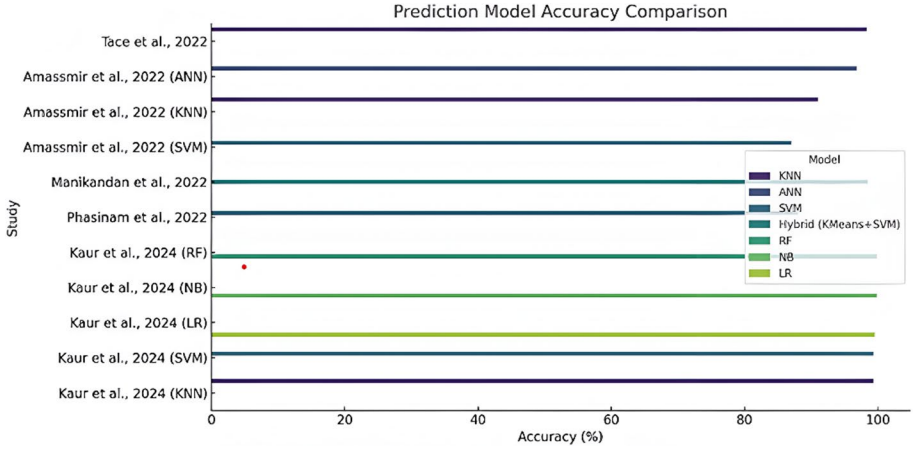


Fig. 7 Comparison of Mean Squared Error (MSE) Values Among ML Models for Irrigation Forecasting. This bar graph compares the MSE values of various prediction models used in smart irrigation. A lower MSE indicates higher prediction accuracy. Kaur et al. [47]. achieved the lowest MSE (0.008) with RF and NB models, suggesting minimal prediction errors. Tace et al. [36] and Manikandan et al. [44] also had low MSE values, while Phasinam et al. [45] and Bhosale et al. (2020) reported higher MSEs (0.10–0.12), indicating relatively less accurate models

than 4G, the 5G network is perfect for smart applications because it offers ultra-fast data transfer, low latency, high connection density, enhanced spectral efficiency, seamless communication, wide coverage, and energy-efficient networking [20].With technologies like LoRa, ZigBee, and WiFi being widely deployed, agri-food supply chain traceability has a lot of potential for the Internet of Things. Meanwhile, 5G and NB-IoT are anticipated to propel agricultural modernization and intelligence. IoT-based agriculture holds great promise for creating novel solutions as technology develops, especially systems that are inexpensive, self-sufficient, energy-efficient, and simple to manage [48]. The 5G network plays a crucial role in revolutionizing smart farming, enabling real-time, high-speed, and low-latency communication for precision agriculture. In the ARETHOU5A project, 5G-IoT technology is integrated with energy reaping and machine learning techniques to develop a prototype end- to-end smart irrigation system. The ultra-reliable and

high-bandwidth capabilities of 5G facilitate the deployment of WSNs, IoT sensors, and UAVs, ensuring efficient water distribution, real-time crop monitoring, and optimized resource utilization. Additionally, 5G connectivity enhances crop growth monitoring, predictive analytics, and food traceability by supporting large-scale data transmission and AI-driven decision-making, thereby transforming the agricultural industry into a more sustainable, efficient, and intelligent ecosystem [49] Fig. 8.

5.2 Potential for automated decision-making and robotics in irrigation

Robots equipped with actuators and sensors are increasingly used for irrigation, soil moisture assessment, and other agricultural tasks like weeding and animal deterrence. Their mobility allows precise water delivery and environmental monitoring. A coverage path planning algorithm enhances navigation, while a Robot Operating System (ROS) structures control into three layers: sensor data collection, communication, and path planning/decision-making. To ensure sustainability and autonomy, robots can be solar-powered, improving efficiency and reducing environmental impact [50]. Several robots integrate sensors and actuators to perform soil moisture sensing, irrigation, and other tasks. Proximity sensors prevent collisions, and wireless robots can spray water while navigating fields. ROS-based systems manage the overall control, and linear shower systems enable efficient irrigation. Solar power further improves sustainability [51]. The Plant Irrigation Mobile Robot addresses limitations of traditional sprinklers by offering portability, efficiency, and minimal disruption. It uses ZigBee-based wireless operation, reducing space consumption and maintenance needs. Evaluations showed improved watering times, robot speed, and soil moisture levels. Future improvements include solar power, remote web-based control, wireless cameras for plant detection, ultrasonic sensors for obstacle avoidance, and remote sensing integration for precise irrigation [52]. Industry 5.0 is emerging alongside Industry 4.0, emphasizing human–robot

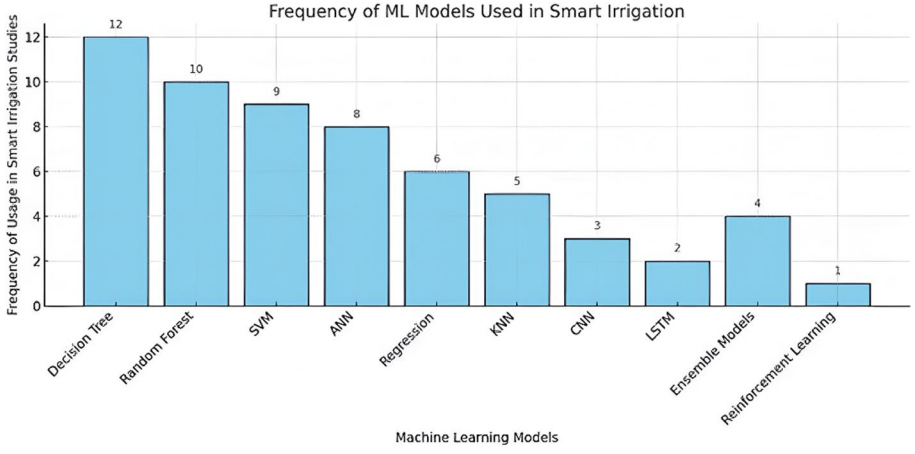


Fig. 8 Frequency of Machine Learning Models Used in Smart Irrigation Research Studies. The bar chart illustrates the frequency of various machine learning models applied in smart irrigation research studies. Among the models, the Decision Tree algorithm is the most utilized, appearing in 12 studies, followed closely by Random Forest (10 studies) and Support Vector Machine (SVM) with 9 instances. Artificial Neural Networks (ANN) also feature prominently with 8 studies. Regression-based models and K-Nearest Neighbors (KNN) were used in 6 and 5 studies, respectively. More advanced models such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Reinforcement Learning appear less frequently, suggesting a developing trend in their application. Ensemble Models were cited in 4 studies, indicating a moderate level of integration. This distribution reflects the growing interest in interpretable and traditional models while hinting at an emerging adoption of deep learning techniques in precision agriculture

collaboration and bridging the physical and digital worlds. It combines human creativity with robotic efficiency through co-working, where humans focus on creative tasks while robots handle repetitive operations. The bioeconomy vision uses biological resources to maintain ecological and economic balance. Industry 5.0 also introduces challenges, including ethics, workplace acceptance, education, and restructuring. Society 5.0, a human-centered digital society, integrates IoT, AI, robotics, big data, and blockchain to address societal challenges. In agriculture, it automates tasks like plowing and sowing while keeping humans in supervisory roles. Society has evolved from hunting-based systems (Society 1.0) to the digital era (Society 5.0), prioritizing human security and well-being [53].With continuous technological progress, robotic arms are expected to play a vital role in future agriculture, enhancing efficiency, innovation, and sustainability [54].

Despite demonstrated potential, IoT-based irrigation systems face socio-economic challenges, especially for smallholder farmers. Initial capital investments for sensors, controllers, and communication infrastructure remain prohibitive, with limited access to affordable financing. Maintenance requirements, including sensor calibration, battery replacement, and connectivity troubleshooting, add to operational burdens. Moreover, low levels of digital literacy hinder effective system use, as many farmers lack familiarity with mobile apps, dashboards, and data interpretation. Capacity-building initiatives, cooperative models, and government subsidies can help overcome these barriers, ensuring broader adoption and sustainable integration of IoT technologies in irrigation.

6 Research gaps

Despite the notable advancements in AI and IoT-enabled smart irrigation systems, several critical research gaps remain. First, standardization across platforms and sensor types is lacking, resulting in interoperability issues and inconsistent performance metrics. Many studies use different sensors, data types, and machine learning algorithms, making it difficult to establish benchmark performance across applications. Moreover, real-time dynamic adaptation of irrigation based on changing environmental factors (e.g., sudden rain) is underexplored. Some systems still rely on semi-manual rule-based models instead of fully automated adaptive models. Furthermore, scalability and energy efficiency for large-field deployments require more attention, especially in resource-constrained rural areas. Lastly, long-term field validation and economic analysis (cost–benefit) are absent in most studies, which is crucial for practical adoption by farmers at scale.

6.1 Standardized benchmarks and federated learning for scalability

One major limitation in current research is the absence of standardized benchmarks and open datasets that allow fair comparison across different IoT-based irrigation studies. Most experiments rely on small, local datasets collected under varying environmental conditions, which restricts reproducibility and scalability. The adoption of federated learning frameworks could help overcome this challenge by enabling collaborative model training across multiple farms or regions without compromising data privacy. This approach would ensure diverse, large-scale datasets while addressing farmers' concerns regarding data sharing and ownership.

6.2 Interoperability and cybersecurity concerns

Interoperability remains a critical bottleneck, as sensor devices, communication protocols (e.g., ZigBee, LoRa, WiFi), and cloud platforms often lack seamless integration. This results in fragmented systems that hinder scalability. Additionally, cybersecurity vulnerabilities—such as unauthorized access to irrigation controllers or data tamper- ing—pose significant risks to adoption. Future research must explore lightweight encryption techniques, blockchain- based data integrity solutions, and standardized IoT frameworks to enhance system trustworthiness.

6.3 Edge–cloud collaboration for real-time and cost-efficient deployment

Another underexplored area is the integration of edge–cloud collaboration architectures. While edge computing reduces latency by processing data closer to the field, cloud platforms are necessary for large-scale analytics and long-term predictions. An optimal balance between edge and cloud resources could minimize costs while maintaining real-time responsiveness. Hybrid frameworks that dynamically allocate tasks (e.g., real-time irrigation control at the edge, predictive modeling in the cloud) represent a promising direction for resource-efficient smart irrigation.

7 Discussion

The reviewed literature collectively underscores the transformation of traditional irrigation systems into intelligent, automated platforms driven by AI, ML, and IoT. The integration of low-cost sensors, wireless data transmission, and cloud platforms such as ThingSpeak has enabled real-time irrigation scheduling. Among ML models, Random Forest and Artificial Neural Networks frequently outperform others, with accuracies extending up to 99.8%. Many researchers are now emphasizing hybrid models that combine ontology (knowledge-based) decision-making and ML for enhanced precision. One common thread is the goal of optimizing water-use with many systems reporting water savings of 37% to 70%. However, these successes often occur in small-scale, lab-controlled environments, and thus the real-world scalability and adaptability remain a challenge. Furthermore, while mobile apps and SMS alerts bridge tech gaps, user-friendly interfaces are needed for low-literate farmers. Future research should explore federated learning for data privacy across regions, renewable energy-powered irrigation systems, and the use of edge computing to reduce latency in real-time decision-making.

7.1 Ethical considerations and data privacy

The adoption of IoT in agriculture raises significant ethical considerations, particularly regarding data privacy and ownership. Farmers generate large volumes of sensor and satellite data, and uncertainties often exist about who controls and benefits from this information. Without adequate safeguards, there is a risk of exploitation by technology providers or third parties. Data anonymization, consent-based sharing, and blockchain-enabled security protocols are emerging as potential solutions. Additionally, ethical dilemmas arise in balancing efficiency gains against social equity, as smallholder farmers may be disproportionately excluded if policies do not ensure fair access and data protection frameworks.

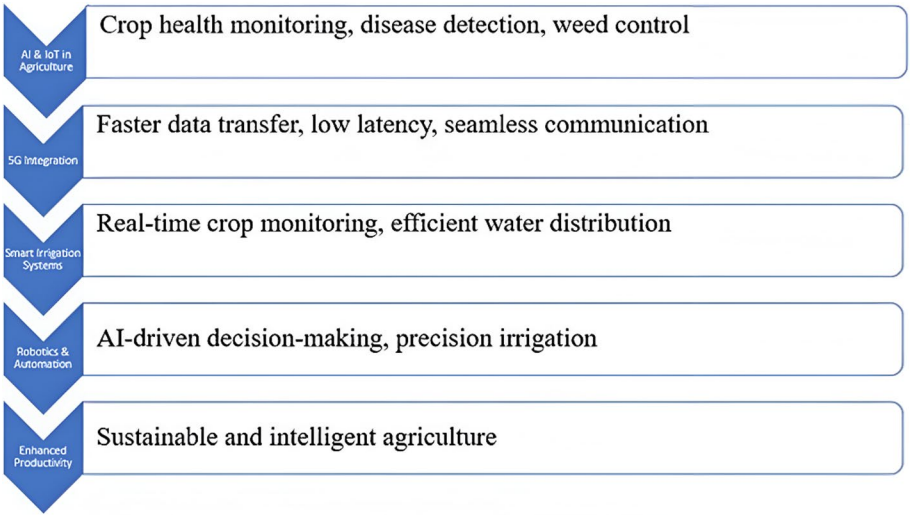


Fig. 9 Potential Aspects and Insights of IOT in Agriculture

7.2 Scalability discussion

Although IoT-based drip irrigation has shown promising results at the pilot and farm scales, scalability remains a critical consideration. Large-scale adoption depends on factors such as interoperability of devices, cloud-based data integration, and affordability for smallholders. Studies suggest that modular IoT frameworks can be progressively scaled from individual plots to community-level irrigation networks—by using standardized communication protocols and edge-cloud integration. However, scalability also raises challenges of data overload, maintenance of distributed sensors, and ensuring equity of access across farm sizes. Addressing these challenges is key for transitioning from isolated demonstrations to large-scale agricultural transformation.

7.3 Future framework proposals

Future IoT-driven drip irrigation frameworks should move toward fully autonomous, data-driven decision support systems. Integration of AI/ML models with IoT sensor networks can provide predictive irrigation scheduling rather than reactive control. Hybrid cloud-edge architectures can enable real-time analytics with lower latency and reduced infrastructure costs. In addition, embedding sustainability metrics (water savings, carbon footprint, energy use) into the irrigation logic can align agricultural practices with climate-resilient goals. A policy-supported and farmer- inclusive framework that ensures affordability, interoperability, and scalability will be essential to translate research into impactful practice Fig. 9.

graphicx

8 Conclusion

Drip irrigation, combined with IoT and precision farming technologies, represents a transformative approach to sustainable agriculture in India. Although there are financial and technical barriers, long-term benefits have increased crop productivity, water conservation, and environmental sustainability, making it an essential solution to address the challenges of climate change and food security. The integration of AI, 5G, and IoT in smart agriculture is further revolutionizing agricultural practices by enabling real-time

crop monitoring, precision irrigation, disease detection, and automated decision-making. AI-powered AgroBots and IoT sensors optimize soil management, reducing manual labor and enhancing resource utilization, while 5G's ultrafast speeds, low latency, and high connection density enable real-time agricultural operations. Advanced robotic systems equipped with AI, sensors and Robot Operating Systems (ROS) facilitate autonomous navigation, precision irrigation, and soil health assessment using Coverage Path Planning (CPP) algorithms. Innovations like ZigBee-based Plant Irrigation Mobile Robots and projects like AREThOU5A showcase the potential of AIoT-driven irrigation to improve efficiency and sustainability. Despite challenges such as high initial investment, technical complexity, and connectivity limitations, investments in affordable IoT solutions, AI-driven automation, and renewable energy integration will be crucial for long-term agricultural resilience and productivity. Studies highlight AI-driven advisory systems, IoT-enabled real-time monitoring, and machine learning models that optimize irrigation efficiency, with some predictive models achieving over 98% accuracy. Microsoft's AI-driven initiative in Andhra Pradesh demonstrated a 30% yield increase, while IoT-based systems using sensors and cloud platforms achieved up to 70% water savings. As technological advancements continue, addressing challenges through cost-effective solutions, farmer training, and improved infrastructure will be key to maximizing the potential of AIoT in sustainable agriculture. The transition towards Industry 5.0, emphasizing human–machine collaboration, will further enhance precision farming, ensuring food security and environmental sustainability in the face of climate change.

Author contributions
N.J. conducted the literature review, conceptualized the structure of the manuscript, and wrote the initial draft. T.V.K. supervised the work, contributed to methodological insights, and critically revised the manuscript. C.S. supported data curation, contributed to the comparison of machine learning models, and assisted with formatting and references. All authors reviewed and approved the final manuscript.

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Data availability
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Declarations

Ethics approval and consent to participate
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